ME/CFS Data Explorer — Project Plan

ME/CFS Data Explorer: A Python + Deep Learning + FastAPI Practice Project
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ME/CFS Data Explorer: A Python + Deep Learning + FastAPI Practice Project

Why this is a good practice project

ME/CFS is scientifically messy on purpose here: no validated biomarker, small cohorts, heterogeneous data types (metabolomics, transcriptomics, questionnaires, wearables). That messiness is exactly what makes it good practice — you'll deal with real-world problems like small-N statistics, missing data, class imbalance, and the temptation to over-claim from a model, rather than a clean Kaggle CSV. It also means every research question below is *genuinely open*, not textbook.

The shape of the project: an end-to-end pipeline — public dataset → cleaning/EDA notebook → PyTorch (or TensorFlow) model → FastAPI service that serves predictions and EDA summaries → optional lightweight dashboard. Small enough to finish, layered enough to touch every skill you listed.

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1. Datasets (real, public, verified)

Dataset	Type	Size	Access	Good for
GSE128078 (PRJNA526259) — whole-blood RNA-seq, ME/CFS patients before/after exercise challenge	Transcriptomics (time series)	14 cases, 11 controls, longitudinal	Free via NCBI GEO	Time-series deep learning practice, differential expression
GSE59489 — DNA methylation (Illumina 450K) in PBMCs	Epigenomics	Case/control	GEO	High-dimensional feature reduction (PCA/autoencoders)
GSE93266 — DNA methylome, larger ME/CFS cohort	Epigenomics	Larger cohort	GEO	Same as above, bigger N
GSE14577 — PBMC gene expression, post-viral CFS vs. healthy	Transcriptomics	8 cases, 7 controls	GEO	Small-N classification, careful cross-validation practice
GSE16059 — monozygotic twins discordant for chronic fatigue	Transcriptomics	44 twin pairs	GEO	Paired-sample statistics, negative-result case study (this study found *no* significant difference — good for teaching that null results matter)
Metabolomics Workbench, Lipkin/Columbia study — plasma metabolomics + fecal metagenomics + clinical data	Metabolomics/microbiome	50 cases, 50 controls	Metabolomics Workbench (search "ME/CFS Lipkin")	Multi-omics fusion, tabular deep learning
Che et al. 2022 plasma metabolomics ("Metabolomic evidence for peroxisomal dysfunction")	Metabolomics	106 cases, 91 controls, 888 analytes, 5 US sites	Supplementary data via PMC9320121	The best-powered public ME/CFS omics dataset — good primary dataset for your classifier
fMRI + n-back memory task dataset (Provenzano et al. 2020)	Neuroimaging	38 CFS, 31 controls	Described in PMC7000378; request via Georgetown/Baraniuk Lab	Stretch goal — imaging pipelines, transfer learning
Synthetic ME/CFS questionnaire generator (Lacasa et al. 2023, Vall d'Hebron)	Synthetic tabular (SF-36, HAD, SCL-90R, FIS, PSQI)	Trained on 2,522 real patients	Paper: Scientific Reports — check supplementary materials/corresponding author for the generator itself	Because it's synthetic, it's a safe first dataset for building your pipeline before touching real patient data
CDC NHANES (general population, has fatigue/sleep/functioning survey items)	Tabular survey	Tens of thousands	NHANES data portal — no ME/CFS diagnosis field, but has self-reported fatigue and functional-limitation items	Practicing FastAPI + real government tabular data at scale before moving to the small clinical datasets

A practical note on access: GEO datasets download as raw expression matrices (`.txt` / `.soft` / `.CEL` files) via `GEOparse` or direct FTP — that's real practice in itself. The omics datasets are small in *sample count* (rows) but huge in *feature count* (columns: hundreds to thousands of genes/metabolites), which is a distinct ML problem from "big data" — you'll need dimensionality reduction, regularization, and honest small-N cross-validation (leave-one-out or repeated stratified k-fold, not a single train/test split).

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2. Research questions to frame your work around

Pick one as your spine; treat the rest as extensions.

1. **Classification under small N:** Can plasma metabolomic/lipidomic profiles (Che et al. dataset) separate ME/CFS cases from controls better than a simple logistic regression baseline — and does a neural net actually help here, or does it overfit? (This is a genuinely useful question — the original paper got AUC ~0.87 with classical ML; beating or replicating that with a small MLP in PyTorch is a real, calibrated goal.)
2. **Subgroup structure:** Within ME/CFS cases, do unsupervised methods (clustering, autoencoders + k-means) recover meaningful subtypes (e.g., by IBS comorbidity, sex, symptom severity) rather than a single homogeneous "ME/CFS" class?
3. **Exercise-challenge signal:** In the longitudinal RNA-seq data (GSE128078), does gene expression actually shift after exertion in a way a model can detect — and how does that compare to the original paper's negative finding? (Reproducing a null result honestly is a great exercise in resisting p-hacking.)
4. **Twin discordance:** With the monozygotic twin dataset, what does a paired/siamese-network approach do differently from a standard classifier, given the original study found no significant transcriptomic difference?
5. **Synthetic-to-real transfer:** Train a model on the synthetic questionnaire data, then discuss (you likely won't have real patient questionnaire data to test on) what would need to hold for that model to generalize — a good essay/writeup exercise in dataset shift.
6. **Feature importance vs. biological plausibility:** Whatever model you build, use SHAP or integrated gradients to extract top features, then check them against the literature (e.g., do your top metabolites match plasmalogens/sphingomyelins/phosphatidylcholines, which multiple independent studies flag?). This teaches you to sanity-check a model against domain knowledge instead of trusting it blindly.

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3. Suggested architecture

```
mecfs-explorer/
├── data/
│   ├── raw/              # downloaded GEO/Metabolomics Workbench files (gitignored)
│   ├── processed/        # cleaned parquet/csv
│   └── download_data.py  # scripted fetch (GEOparse, requests)
├── notebooks/
│   ├── 01_eda.ipynb
│   ├── 02_baseline_sklearn.ipynb   # ALWAYS build the boring baseline first
│   └── 03_deep_model.ipynb
├── src/
│   ├── data/              # loaders, cleaning, feature engineering
│   ├── models/             # PyTorch nn.Module or TF model definitions
│   ├── train.py
│   └── evaluate.py          # cross-validation, calibration, SHAP
├── api/
│   ├── main.py              # FastAPI app
│   ├── schemas.py           # Pydantic request/response models
│   └── routers/
│       ├── predict.py        # POST endpoint: features in -> class + confidence out
│       └── eda.py            # GET endpoints serving summary stats/plots as JSON
├── tests/
├── Dockerfile
└── requirements.txt
```

API design worth practicing (this is where FastAPI actually earns its keep, not just as a wrapper):

- `POST /predict` — takes a feature vector (Pydantic model with validation), returns predicted class + probability + top contributing features (SHAP values) — teaches you to *not* return a bare label, since a health-adjacent model should always surface uncertainty and caveats
- `GET /dataset/summary` — serves descriptive stats (means, missingness, group sizes) computed at startup
- `GET /model/metadata` — returns training date, dataset version, performance metrics, and a hard-coded disclaimer that this is not diagnostic
- Background task or `/retrain` endpoint (optional, stretch) — practice with FastAPI's async and background task support

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4. Phased roadmap

Phase 1 — Python + data refresh (few days) Pick the Che et al. metabolomics dataset or NHANES as your warm-up. Do EDA in pandas: missingness, class balance, distribution plots. Build a scikit-learn baseline (logistic regression, random forest) with proper stratified cross-validation. This baseline is your benchmark for everything after — if your neural net can't beat it, that's a real and useful finding, not a failure.

Phase 2 — Deep learning (1–2 weeks) Build a small MLP in PyTorch (or TensorFlow/Keras) for the same tabular task. Practice: custom `Dataset` / `DataLoader`, handling class imbalance (weighted loss or SMOTE), regularization (dropout, weight decay) since N is small, early stopping, and k-fold cross-validation loops (not just one split — small-N deep learning without this is close to meaningless). Add SHAP or captum for explainability.

Phase 3 — FastAPI service (3–5 days) Wrap the trained model behind the API design above. Practice Pydantic validation, dependency injection (e.g., loading the model once at startup), async endpoints, error handling, and OpenAPI docs (`/docs` comes free — inspect it). Write a few `pytest` tests with `TestClient`.

Phase 4 — Stretch goals (pick what excites you) - Dockerize the API and deploy it (Fly.io / Render free tier) so it's a live, shareable artifact - Add the RNA-seq time-series dataset and try a small RNN/1D-CNN for the exercise-challenge question - Add a lightweight frontend (Streamlit, or a small HTML page hitting your API) for interactive EDA - Write up your findings against the published literature — did your feature importances match?

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5. Honest caveats to build in from day one

- **N is small everywhere.** A model reporting 95% accuracy on 90 samples is almost always overfit — treat any single high number with suspicion and always report cross-validated ranges, not point estimates.
- **No dataset here is a diagnostic ground truth.** ME/CFS diagnosis itself is criteria-based (e.g., Institute of Medicine / Canadian Consensus Criteria), not lab-confirmed, so "case" vs. "control" labels carry real uncertainty. Frame your model's job as "distinguishing this study's labeled cohorts," not "diagnosing ME/CFS."
- **Some data needs a data use agreement or institutional request** (fMRI dataset, some Metabolomics Workbench raw files) — GEO and NHANES are the easiest fully-open starting points.
- If you ever write this up publicly, it's worth a line of respect for the patient community: this disease has a long history of being dismissed, and "a hobby project found X" headlines can do real harm if overstated.